

Airline Delay and Operations Intelligence

Houston, 2011 — 227,496 flights

DSA 3050A — Business Intelligence & Visualization · Grp05

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The business problem

Houston Airport System — Operations Performance

- Delay is the largest controllable cost in airport operations — fuel, crew duty limits, gate reallocation, missed connections, compensation.
- Houston ran 227,496 departures in 2011. 47.8% of operated flights arrived late.
- Management could see *that* the network was late — not *where the lateness was being created*.
- Without attribution, mitigation spending is guesswork.

Our brief: attribute delay and cancellation to the carrier, route, airport, hour and month that produced them.

The data

hflights — Houston departures, 2011

227,496	21 →	15	149	116
	32			
FLIGHTS	COLUMNS	AIRLINES	ROUTES	DESTINATIONS

- **Source:** US DOT Bureau of Transportation Statistics, via the R *hflights* package — genuine public government reporting.
- **Period:** every commercial departure from IAH and HOU in 2011.
- **Structurally clean:** zero duplicates, zero blank rows — but times stored as HHMM integers, flags as 0/1, and a 98.69% null rate on cancellation code.

Cleaning and modelling

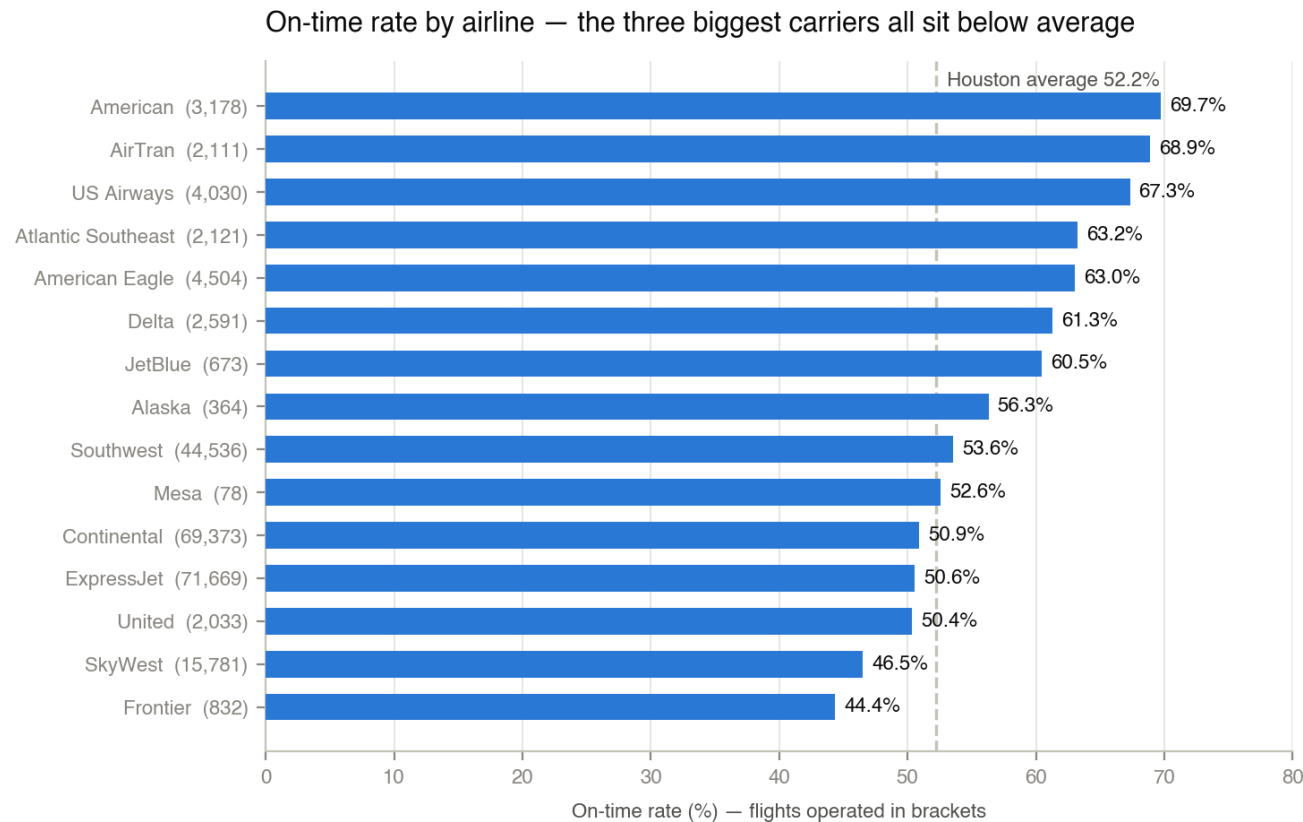
One fact table, three dimensions, 22 measures

All 10 required cleaning tasks, and 8 advanced tasks against a requirement of 6.

- Renamed 13 columns to business language; decoded every code and flag.
- Split HHMM integers into real clock times; merged origin and destination into a Route key.
- Derived Flight Status, Delay Category, Distance Band, Time of Day, Season and Delay Recovered.
- Star schema: Fact_Flights + Dim_Date + Dim_Airline + a Group By aggregate table, measures isolated in _Measures.

The decision that mattered: 3,622 null arrival delays belong to cancelled and diverted flights. We left them null. Filling them with zero would have counted 3,622 flights that never arrived as "on time".

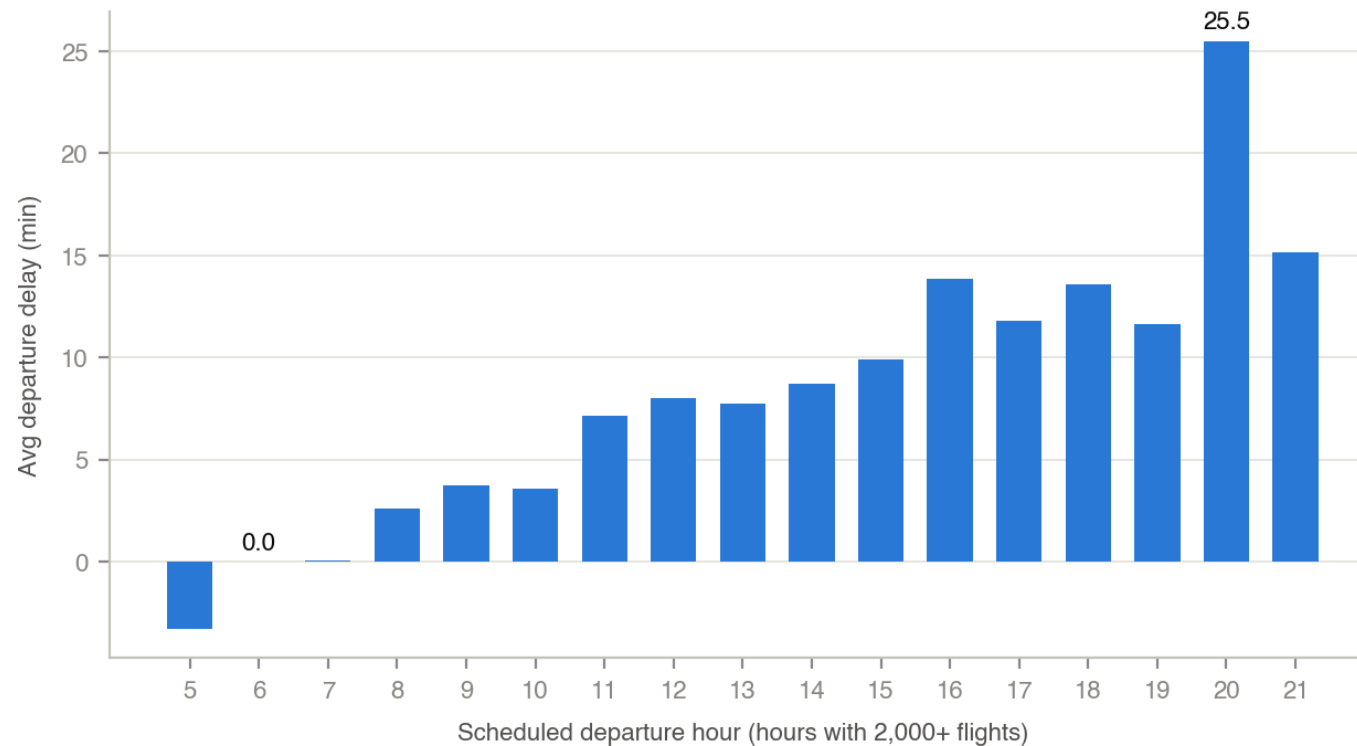
Insight 1 — Punctuality is inversely related to scale



Three carriers fly 82% of Houston. All three run at or below the 52.2% average. Improving the other twelve cannot move the system.

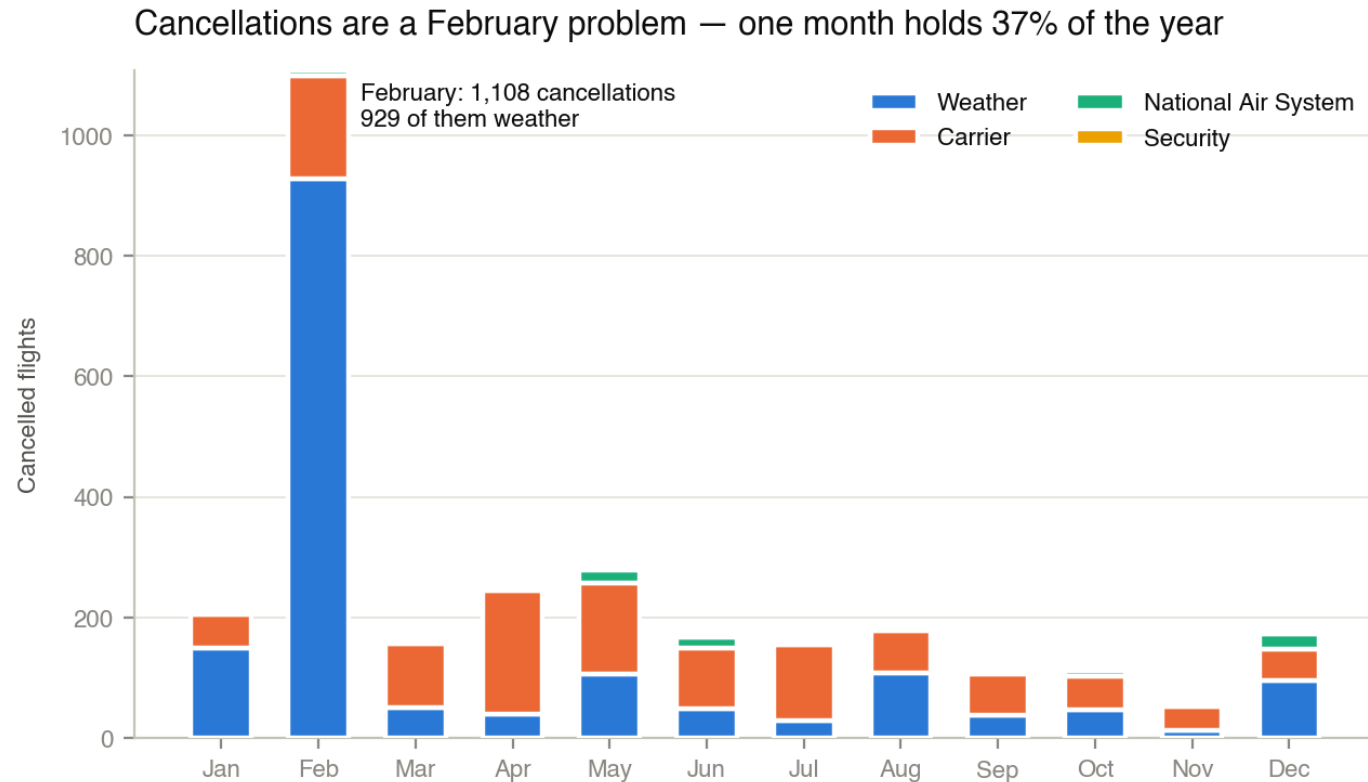
Insight 2 — Delay is manufactured during the day

Delay compounds through the day — a 06:00 departure leaves on time, a 20:00 departure averages 25 minutes late



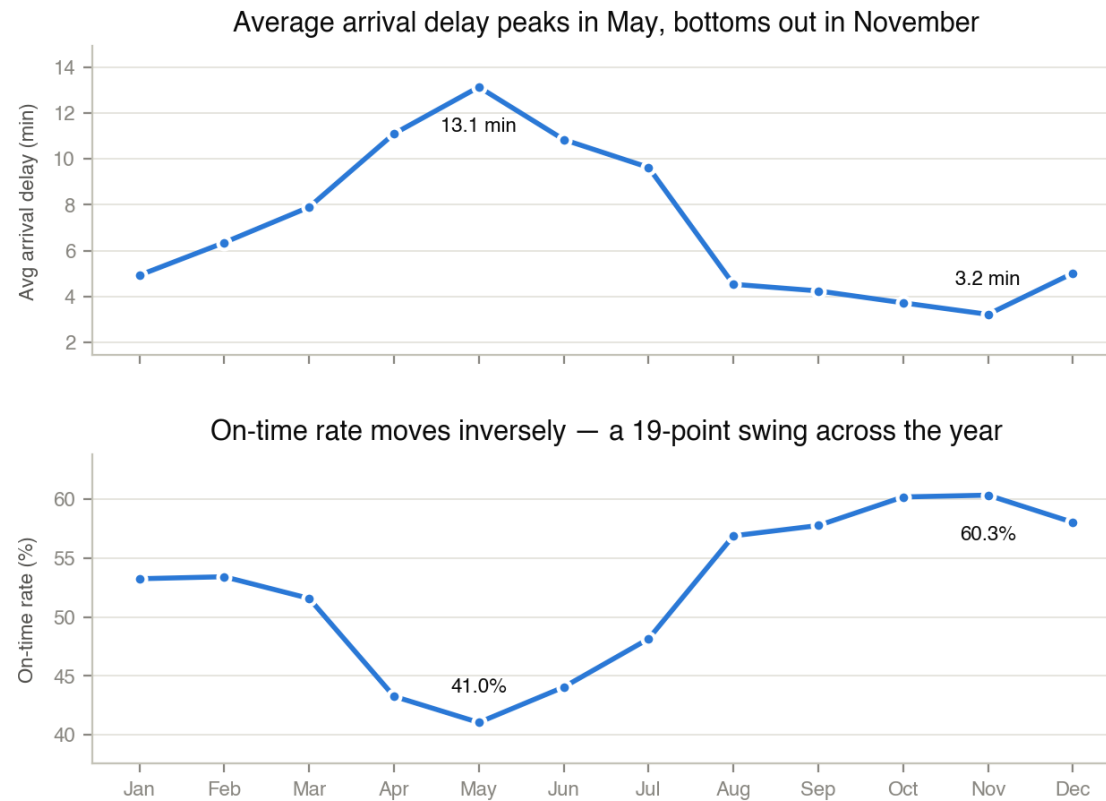
06:00 departs on time. 20:00 averages 25.5 minutes late. Same aircraft, same crews, same routes — delay compounds across rotations.

Insight 3 — Cancellations are one week in February



1,108 of 2,973 cancellations fall in February. Strip it out and the network cancels 0.89% — a well-run operation.

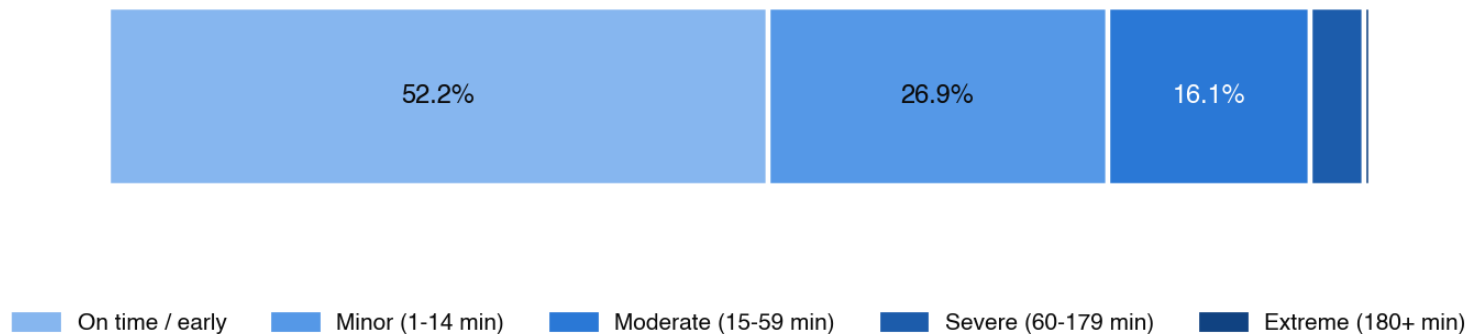
Insight 4 — Delay peaks in May, traffic peaks in July



Volume does not predict delay: $r = 0.33$. Contingency staffing is aligned to the wrong month.

Insight 5 — A thin tail does the damage

Delay severity mix of 223,874 completed flights



79% of flights are effectively fine. The cost sits in the 4.7% — 10,584 flights — that arrive an hour or more late, of which 1,110 are over three hours late. An average-delay target chases the harmless middle.

What we recommend

Six recommendations, traced to the evidence

#	Recommendation	Why
1	Target the three high-volume carriers	Moving them nine points moves the whole system seven. Perfecting the other twelve moves it under three.
2	Protect the morning bank	Delay compounds from 06:00. The earliest minute saved is the cheapest.
3	Fix HOU's turnaround, not its taxiways	Half the taxi time, more departure delay. The gate is the constraint.
4	Separate February from the cancellation baseline	The blended rate hides a good operation and misnames the dominant cause.
5	Target severe-delay rate, not average delay	Aims at the 10,584 flights that break connections, not at noise.
6	Load-plan for May, not July	Align contingency with when the network fails, not when it is busiest.

Where this leaves Houston

- Houston does not have a general punctuality problem. It has three addressable problems and one unavoidable one.
- **Unavoidable:** February. One ice storm, 37% of the year's cancellations.
- **Concentration:** three carriers are the system.
- **Compounding:** delay is created hour by hour, not inherited.
- **Measurement:** the current metrics point management at the least actionable part of all three.

The cheapest recommendation — change what is measured — costs nothing and unlocks the value of the other five.

Thank you

Questions